

Constant-force spring

An ideal **constant-force spring** is a spring for which the force it exerts over its range of motion is a constant, that is, it does not obey Hooke's law. In reality, "constant-force springs" do not provide a truly constant force and are constructed from materials which do obey Hooke's law. Generally, constant-force springs are constructed as a rolled ribbon of spring steel such that the spring is in a rolled up form when relaxed.

As the spring is unrolled, the material coming off the roll (un)bends from the radius of the roll into a straight line between the real and the load. Because the material tension-stiffness of the straight section is orders of magnitude greater than the bending stiffness of the ribbon, the straight section does not stretch significantly, the restoring force comes primarily from the deformation of the portion of the ribbon near the roll. Because the geometry of that region remains nearly constant as the spring unrolls (with material coming off the roll joining the curved section, and material in the curved section joining the straight section), the resulting force is nearly constant.

References

- Ansel C. Ugural, *Mechanical Design of Machine Components, Second Edition*, page 683 (<https://books.google.com/books?id=4pcXBgAAQBAJ&pg=PA683>), CRC Press, 2015 ISBN 1439887810.

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